Example 9.13

Let X be a discrete random variable with the following probability distribution

X	-3	6	9
$P(X=x)$	$\frac{1}{6}$	$\frac{1}{2}$	$\frac{1}{3}$

Find mean and variance

Solution:

$$\text{Mean} = E(X) = \Sigma x_i p_i$$

$$= x_1 p_1 + x_2 p_2 + x_3 p_3$$

$$= -3\times \frac{1}{6} + 6 \times \frac{1}{2} + 9 \times \frac{1}{3}$$

$$= \frac{11}{2}$$

$$E(X^2) = \Sigma x_i^{\,2} p_i$$

$$= x_1^{\,2} p_1 + x_2^{\,2} p_2 + x_3^{\,2} p_3$$

$$= \left(-3^{2\times}\frac{1}{6}\right) + \left(6^{2\times}\frac{1}{2}\right) + \ldots \left(9^{2\times}\frac{1}{3}\right)$$

$$= \frac{3}{2} + 18 + 27$$

$$= \frac{93}{2}$$

$$\text{Van}(X) = E(X^2) - (E(X))^2$$

$$= \frac{93}{2} - \left(\frac{11}{2}\right)^2$$

$$= \frac{93}{2} - \frac{121}{4}$$

$$= \frac{186 - 121}{4}$$

$$= \frac{65}{4}$$

$$\text{Mean} = \frac{11}{2},$$

$$\text{Var}(X) = \frac{65}{4}$$

9.6.2 Expectation of a continuous random variable

Let X be a continuous random variable with probability density function $f(x)$ then the mathematical expectation of X is defined as